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Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21304

F–404

B.C.A. (Ist Semester) Examination, 2019–20

MATHEMATICS–I

(New Course)

Paper No. BCA–104–N

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) Evaluate :

$$\lim_{x\to 0}\frac{x^{1/2}\tan x}{(e^x-1)^{3/2}}$$

(b) Discuss the continuity of $f(x) = [x]$ at $x = 0$, where $[x]$ denotes the integral part of x.

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2. (a) Find n^{th} differential coefficient of $\tan^{-1}\frac{x}{a}$.

(b) If :

$$y^{1/m} + y^{-1/m} = 2x,$$

prove that :

$$(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)\,y_n = 0$$

3. (a) If :

$$u = \tan^{-1}\frac{xy}{\sqrt{1+x^2+y^2}},$$

show that :

$$\frac{\partial^2 u}{\partial x\partial y} = \frac{1}{(1+x^2+y^2)^{3/2}}$$

(b) If :

$$u = \sin^{-1}\frac{x^2+y^2}{x+y},$$

show that :

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$$

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4. (a) If :

$$x = a \cosh \xi \cos \eta$$

$$y = a \sinh \xi \sin \eta$$

prove that :

$$\frac{\partial(x, y)}{\partial(\xi, \eta)} = \frac{a^2}{2}(\cosh 2\xi - \cos 2\eta)$$

5. (a) Evaluate :

$$\int_0^{2a} \frac{x^{a/2}}{\sqrt{2a - x}} dx$$

(b) Show that the ordinate $x = a$ divides the area between $y^2(2a - x) = x^3$ and its asymptote into two parts in the ratio $3\pi - 8$; $3\pi + 8$.

6. (a) Prove that :

$$\nabla^2(r^n \bar{r}) = n(n+3)r^{n-2}\bar{r}$$

(b) Show that :

Curl grad $r^n = 0$

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7. (a) Solve :

$$(1 + x^2)\frac{dy}{dx} + y = \tan^{-1} x$$

(b) Solve :

$$(D^2 + 1)y = 8x^2 e^{2x} \sin 2x$$

8. (a) Obtain the equation of the ellipse whose letus rectum is 5 and whose eccentricity is 2/3, the axis of the ellipse being the axes of coordinates.

(b) Find the eccentricity and co-ordinate of the foci of the curve $3x^2 - y^2 = 4$.

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